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Polymerization of renewable itaconic acid in deep eutectic monomers: Effect of the quaternary ammonium cation structure

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ABSTRACT

A systematic investigation has been carried out on the synthesis of poly(itaconic acid) (PIA) in Deep Eutectic Monomer (DEM) systems, composed of bio-derived itaconic acid (IA) and quaternary ammonium chlorides (QCl) varying in their cation structure. The physicochemical properties including density, viscosity, Kamlet-Taft parameters and polarity as Nile Red transition energy of DEMs supported with spectroscopic studies (FT-IR, 1H and 13C NMR) are reported. It was observed that cation structure of the ammonium chloride impacts on the physicochemical properties and radical polymerization of DEMs. More notably, the correlations between the Kamlet-Taft hydrogen bonding accepting ability (β) of the DEMs and both IA polymerization rate and the molecular weight of the synthesized PIA were found. These findings provided theoretical guidance on designing DEMs tailored to specific polymerization process.

1. Introduction

Itaconic acid (IA) is a vinyl monomer of renewable origin, industrially produced by fermentation of various biomass feedstocks [1]. IA and its derivatives are versatile building blocks suitable for preparation of renewable polymers by both polycondensation and radical polymerization [2,3]. In recent years, IA has been attracting scientific attention for synthesis of superabsorbents [4,5] clay-nanocomposites [6], antibacterial polymers [7,8], and also curable polyesters [9] and their reactive diluents [10]. Due to similarity in chemical structure, IA is considered a potential bio-based alternative for such petrochemical monomers as acrylic and methacrylic acids [11]. However, due to the steric hindrance of the vinyl bond by two carboxyl groups, radical homopolymerization of IA proceeds slowly and results in a low molecular weight polymers [11–14]. Poly(itaconic acid) (PIA) with a molecular weight $< 100\,000$ g/mol is typically obtained in both aqueous and organic solutions [13,14]. However, by using Deep Eutectic Solvent (DES) as an alternative reaction medium for IA polymerization, a high molecular weight PIA ($M_w \approx 900\,000$ g/mol) can be synthesized [15] and thus utilised for preparation of polymer materials, e.g. self-assembled hydrogels [16,17].

DESs are binary or ternary mixtures, formed by hydrogen bonding between the components of the mixture, where the hydrogen bond donors (HBD) are represented by alcohols, amides, carboxylic acids while the hydrogen bond acceptors (HBA) are the quaternary ammonium salts [18,19]. Characteristic feature of DESs is a „deeply" reduced melting point compared to the individual components, mainly due to the formed H-bonds causing charge delocalization [18,19]. Since DESs can be prepared from non-toxic, biodegradable and relatively inexpensive renewable feedstocks, they have attracted much attention as potential sustainable alternatives to commonly used organic solvents [20,21]. Moreover, due to the tunability of their structure, strong potential for commercial use of DES as reaction media for special applications rather than as mere substitutes for organic solvents can be expected [22,23]. Therefore, it seems highly desirable to investigate in detail the relationships between the DES composition and its ability to liquefy substances with high melting points to fully explore the potential of DES as a reaction medium [22,23].

Deep eutectic monomers (DEMs) are defined as DESs with ability to polymerize either through monomeric nature of HBD (e.g. acrylic acid, methacrylic acid, acrylamide) or HBA (e.g. quaternary ammonium vinyl derivatives) [24]. Polymerized DEMs (PDEMs) have recently found many applications e.g. as solid electrolytes and flexible conductive materials [24–27]. Although PDEMs applications have been extensively

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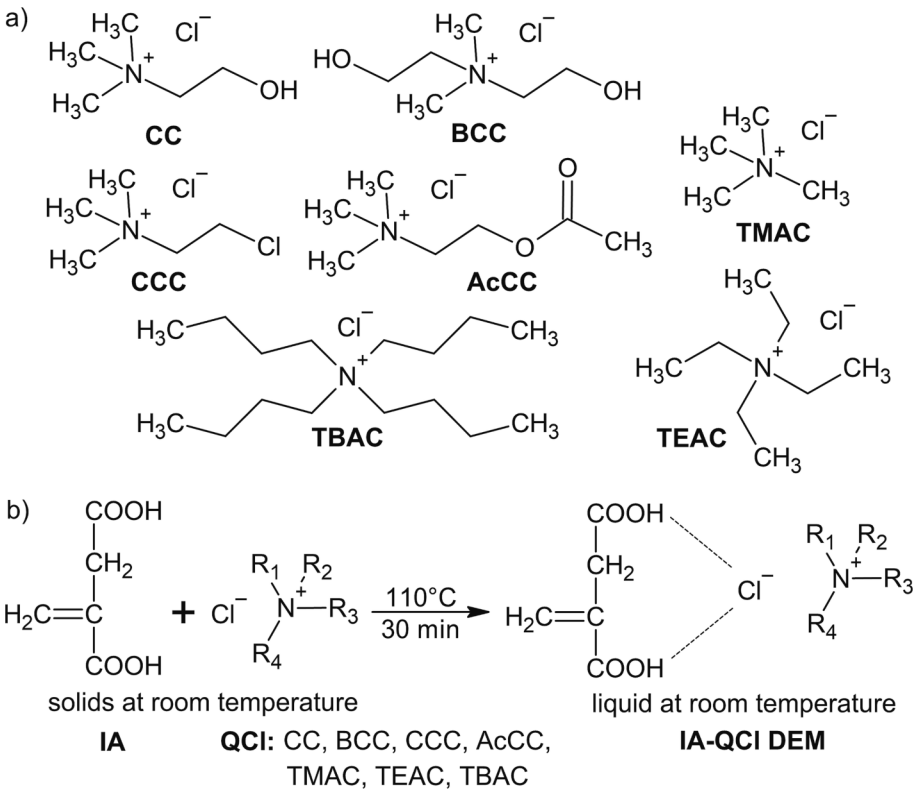


Fig. 1. a) Structures of quaternary ammonium chlorides (QCl) used in this study. b) Preparation of IA-QCl DEMs from IA as HBD and QCl as HBA.

studied, the kinetics of their polymerization has so far been scarcely investigated [28,29]. However, even from these few studies, it is clear that both the rate of polymerization and the molecular weight of the resulting polymers can be largely influenced by the DEM composition [28,29].

IA is one of vinyl monomers known to form DEMs [30]. An equimolar mixture of IA and choline chloride (CC) – IA-CC 1:1 is liquid at room temperature. Moreover, the molar concentration of IA in DEM is higher than that of IA in common solvents (water or alcohols) [31]. IA in IA-CC 1:1 DEM polymerizes much more rapidly than in aqueous solution, which solves the problem of its slow radical polymerization [32]. Simultaneously, we have recently demonstrated that under appropriate reaction conditions i.e. mild temperature or slow initiation rate, it is possible to obtain PIA with a high molecular weight ($M_w \approx 900\,000$ g/mol) using IA-CC 1:1 DEM [15]. Furthermore, our previous study revealed that the type of quaternary ammonium chloride (QCl), which was used for formation of DEM with IA, affected the molecular weight of the resulting PIA [33]. However, it has not been clarified yet how the chemical structure of QCl affects the polymerization of IA, and which type of QCl is the most appropriate for obtaining high molecular weight PIA.

In this work we have studied the effect of cation structure of QCl on the progress of IA radical polymerization. Particularly, quaternary ammonium cations with different alkyl chain length and functional groups (Fig. 1) were selected. Then, the physicochemical properties, namely density, viscosity and polarity (Kamlet-Taft parameters and Nile Red transition energy), of IA-QCl DEMs were investigated to determine the effect of QCl structure on IA polymerization. We hypothesize that the above-mentioned parameters will have the most significant impact on the polymerization process as it is known that: (i) the density of DEMs is a factor, which limits the maximal initial molar concentration of the monomer (IA) in the system, thus affecting the polymerization kinetics [34]; (ii) the viscosity of the reaction system affects the termination reaction rate of radical polymerization [35]; (iii) Kamlet-Taft parameters as well as FT-IR and ^{1}H and ^{13}C NMR spectral data provide information about molecular state of the monomer. These DEMs properties were further correlated with results from polymerization experiments aiming to determine the key parameters for obtaining high molecular weight PIA.

2. Experimental

2.1. Materials

Itaconic acid (IA, ≥99 %), choline chloride (CC, ≥98 %), chlorocholine chloride (CCC, ≥98 %), acetylcholine chloride (AcCC, ≥99 %), tetramethylammonium chloride (TMAC, ≥98 %), tetraethylammonium chloride (TEAC, ≥98 %), tetrabutylammonium chloride (TBAC, ≥97 %), 2,4,6-trimethylbenzoyldiphenyl phosphine oxide (TPO, ≥97 %), Nile Red (NR, ≥99 %), 4-nitroaniline (NA, ≥99 %) were purchased from Sigma-Aldrich. Bis(2-hydroxyethyl)dimethylammonium chloride (BCC, ≥98 %) was purchased from TCI. N,N-diethyl-4-nitroaniline (DENA, ≥98 %) was obtained from Fluorchem. Prior to use solid IA was ground in a mortar with pestle. The rest of the chemicals were used as received.

2.2. Preparation of DEMs

The DEMs were prepared according to methodology described in previous works [30,33] – certain amounts of IA and QCl were pre-mixed in a glass vial, and then melted together by heating for 30 min at 110 °C while stirring. For all tests, the DEMs were freshly prepared.

2.3. Density measurements

The density of the DEMs was measured by a 1 ml pycnometer at room temperature in triplicate and the average values are reported.

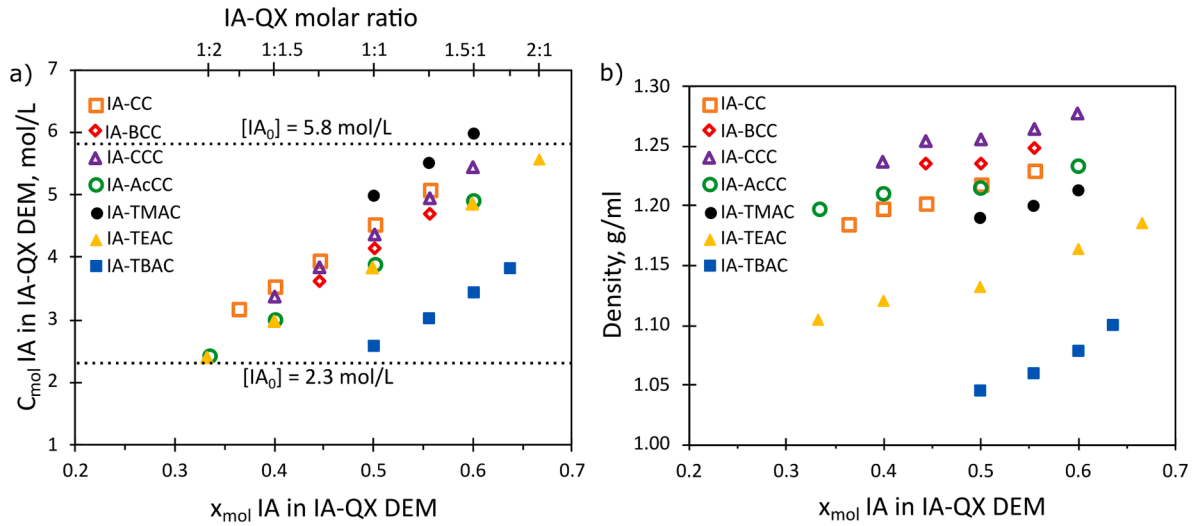


Fig. 2. a) Compositions of the investigated IA-QCl DEMs corresponding to the liquid window at room temperature expressed by different concentration units. The [IA0] values indicate the range of IA initial molar concentrations in which the polymerizations were carried out (Tab. S3). b) Density of IA-QCl DEMs at room temperature.

2.4. Viscosity measurements

The viscosity of DEMs was examined using a rotational rheometer MCR 302 (Anton Paar GmbH, Austria) with plate-plate geometry (diameter of 25 mm). The shear rheological experiments were performed in the shear rate range of 10^1 to 10^2 s^{-1}. Measurements were conducted at 25 °C. The temperature was maintained using the integrated thermostatic system.

2.5. Determination of solvatochromic parameters

Small amount (c.a. 0.1 mg) of each solvatochromic probe (NR, DENA and NA) was added to the investigated DEMs (ca. 1 ml) and mixed at 110 °C by magnetic stir bar. After cooling down to room temperature UV–Vis spectra (UV-2600 Shimadzu) of the DEM with the respective dye was recorded to determine the probe maximum absorption wavelength (λ_{max}). The solvatochromic parameters: The molar transition energy of Nile Red (E_{NR}), hydrogen-bond accepting ability (β) and polarisability/dipolarity parameter (π*), were calculated using the following Eqs. (1)–(3) [36]:

$$E_{NR} = 28591/\lambda_{NR} \quad (1)$$

$$\beta = (1.035/\lambda_{maxDENA} + 2.64 - 1/\lambda_{maxNA})/2.8 \quad (2)$$

$$\pi^* = 0.314(27.52 - 1/\lambda_{maxDENA}) \quad (3)$$

The spectra of NR in DEMs were deconvoluted using open source *Fityk* software [37]. The NR spectra, which were found to be multimodal, are discussed in Supplementary material (Fig. S1). The E_{NR} was determined on the basis of λ_{max} of the monomeric form of NR [38].

2.6. Fourier-transform infrared spectroscopy

The FT-IR analyses were performed at room temperature using IRSpirit FT-IR (Shimadzu, Japan) spectrometer equipped with a crystal in horizontal attenuated total reflectance (ATR) cell. Spectra were acquired in a range from 4000 to 400 cm^{-1} with 1 cm^{-1} resolution and average of 16 scans.

2.7. Nuclear magnetic resonance spectroscopy

In-situ high-resolution ^1H and ^{13}C NMR spectra of IA-QX DEMs were acquired with Bruker Avance III 600 spectrometer operating at 600.2 MHz, and 150.9 MHz respectively. For ^1H NMR the width of 90° pulse was 18 μs, relaxation delay 15 s, acquisition time 1.82 s, 2 scans. For ^{13}C NMR the width of 90° pulse was 14 μs, relaxation delay 10 s, acquisition time 0.77 s, 16 scans. The frequency of the spectrometer was not locked during data acquisition due to the lack of a deuterated solvent. The magnetic field was shimmed manually for each sample to minimize magnetic field inhomogeneity. All ^1H and ^{13}C chemical shifts were externally referenced to the deuterated DMSO signal shortly before the spectra were collected [39].

2.8. Photopolymerization monitored by real-time FT-IR

The progress of free-radical photopolymerization of IA in IA-QCl DEMs was followed by real-time FT-IR spectroscopy using FT-IR i10 NICOLETTM spectrometer equipped with a horizontal adapter (Thermo Scientific, USA). The evaluation of vinyl group content was continuously followed at about 6164 cm^{-1} (vinyl C═C vibration overtone). The monomer conversion was calculated by using Eq. (4):

$$X_{IA}[\%] = (1 - A_t/A_0) \bullet 100\% \quad (4)$$

where: A_0 is the area of the absorbance peak of C═C vibration overtone before polymerization, and A_t is the area of the peak at certain photopolymerization time. LED emitting the blue edge of the visible spectrum (emission maximum 405 nm, Thorlabs Inc., USA) with the intensity 0.4 mW/cm^2 was used to initiate the photopolymerization. The light source was switched on 10 s after the start of spectra acquisition. For photopolymerization process, DEMs were placed in a 1.4 mm thick ring at a distance of 2.1 cm from the LED. The initial concentration of the TPO photo-initiator in each reaction system (DEM) was 0.0045 mol/L and photopolymerization was carried out for 30 min. Additionally, during the photopolymerization process, the surface temperature of the sample was monitored using a thermal imaging camera (Dongguan Xintai Instrument Co., Ltd). The initial polymerization rates (R_p) were determined from the slope of linear course of conversion curves in the initial phase of the polymerization reaction.

2.9. Size exclusion chromatography of polymers

Polymers (PIA) were isolated from polymerized DEMs by dialysis, followed by freeze-drying. Size exclusion chromatography (SEC) was carried out in phosphate buffer (0.1 M Na_2HPO_4) as the eluent at 40 °C

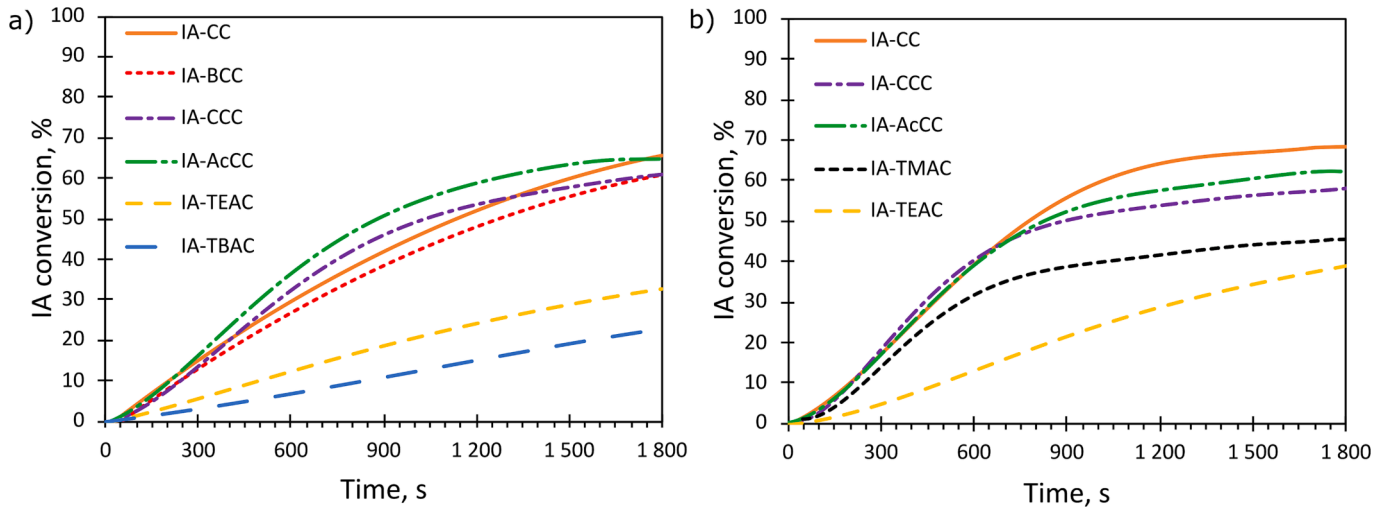


Fig. 3. Photopolymerization IA conversion curves in IA-QCl DEMs for a) $[IA_0] = 3.8$ mol/L, b) $[IA_0] = 4.8$ mol/L.

using Knauer AZURA liquid chromatography system (KNAUER, Germany) with Knauer Azura RID 2.1L refractive index detector. Analyses were conducted using a PSS SUPREMA XL Linear column calibrated with poly(methacrylic acid) sodium salt standards with narrow molecular weight. Samples for analysis were prepared by dissolving 5 mg of PIA in 1 ml of phosphate buffer at room temperature. In addition, PIA obtained from IA-BCC DEMs were hydrolyzed (24 h, 50 °C) in phosphate buffer before analysis, in order to dissolve the polymer by cleavage of cross-linking ester bonds.

3. Results and discussion

3.1. Compositions of IA-QCl DEMs

Preliminary studies (Tab. S1) have shown that among quaternary ammonium salts with various anions (Cl^-, Br^-, I^-, BF_4^-, PF_6^-), only chlorides form binary mixtures with IA, which remain liquid at room temperature. Therefore, DEMs formed from QCls with various cations (Fig. 1) were selected for further studies. HBD-HBA interactions play major role in the disruption of the H-bonded crystal structure of IA [40] resulting in the replacement of IA - IA interactions by IA - QCl complexes in DEMs [41,42]. Knowing that among the anions investigated (Cl^-, Br^-, I^-, BF_4^-, PF_6^-), chloride has the strongest hydrogen bond accepting ability [29,43] it is clear why QCl with IA could form DEMs.

It is well known that to obtain room temperature-liquid DESs, it is necessary to set the ratio of individual components. In this study, we have determined ranges of concentrations – the so-called *liquid window* [44] - at which IA forms liquid DEMs with QCl (Fig. 2). Fig. 2a shows that the DEMs can be prepared using different IA-QCl molar ratios ranging from 2:1 to 1:2 depending on the cation structure used. Specifically, it was found that all IA-QCl DEMs are liquid at room temperature at the equimolar ratio (i.e. $x_{mol\ IA} = 0.5$). As a consequence of using different cations and compositions, the DEMs exhibit densities ranging from 1.05 to 1.28 g/mL (Fig. 2b and Tab. S2). It means that the liquid window corresponds to molar IA concentrations from 2.3 to 5.8 mol/L with implications when planning comparative polymerization studies.

Fig. 2b also shows that the density of the DEMs increases proportionally with increasing IA content, since the pure solid IA (1.573 g/mL at 25 °C) has higher density than QCls. The effect of the type of QCl used on IA-QCl DEMs density is particularly significant for the series of cations differing in alkyl chain length (TMAC, TEAC, TBAC). As the alkyl chain length increases, the density of DEMs decreases (IA-TMAC > IA-TEAC > IA-TBAC), as longer alkyl chains provide more free volume consequently reducing the DEM density [45].

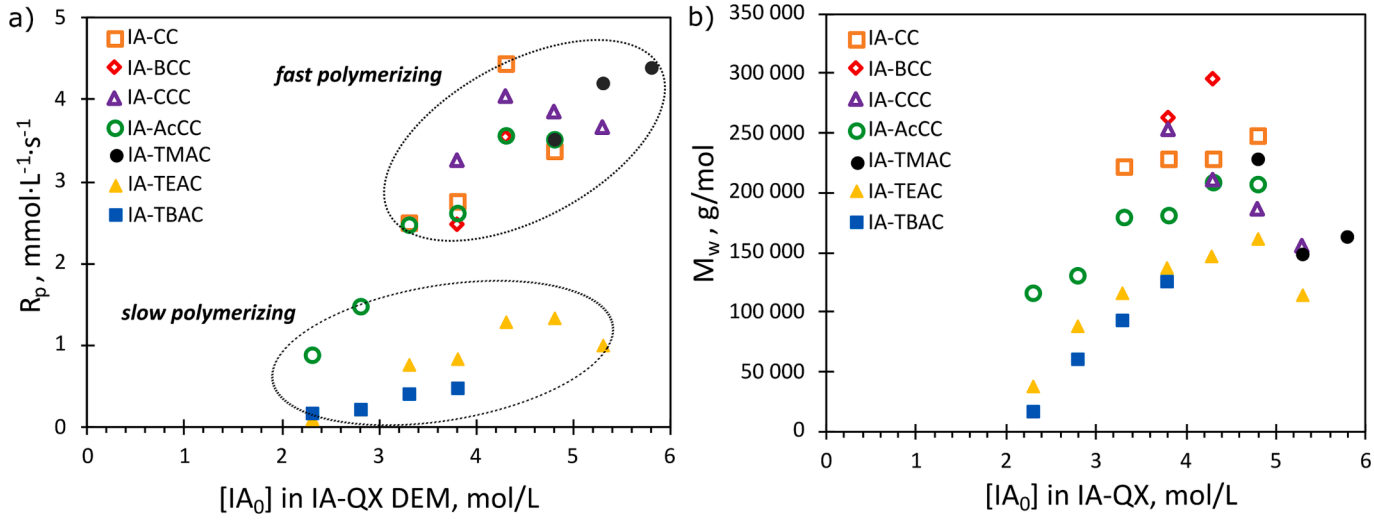


Fig. 4. a) Initial polymerization rate r_p in IA-QCl DEMs as a function of initial IA molar concentration [IA_0] and conventional division of DEMs into two groups: *fast polymerizing* and *slow polymerizing*. b) Weight average molecular weight (M_w) of PIA obtained in IA-QCl DEMs as a function of initial IA concentration.

3.2. Photopolymerization of IA-QCl DEMs

Radical polymerization of DEMs was performed at mild temperature (30–35 °C - Fig. S3) which is crucial for obtaining high molecular weight PIA. Moreover, to reduce heating of DEM mixture during the progress of the reaction, photopolymerization with a low concentration of photoinitiator (TPO) and a low light irradiation intensity was applied.

For each DEM, the polymerization was carried out over the entire range of initial IA molar concentrations ($[IA_0]$ = 2.3 – 5.8 mol/L, see *liquid window* Fig. 2) to assess the effect of the type of QCl on the radical polymerization process. The polymerization results, i.e. initial polymerization rate, final IA conversion; M_n, M_w and $Đ$ of the obtained PIA are summarized in Tab. S3. Due to the significant variation in the liquid window of individual IA-QCl DEMs, it is not possible to compare the polymerization of all DEMs at one selected $[IA_0]$. For this reason, two initial IA concentrations (3.8 and 4.8 mol/L) covering all system were selected (Fig. 3).

Fig. 3a shows the IA conversion curves for individual IA-QCl DEMs photopolymerization at $[IA_0]$ = 3.8 mol/L. It is clearly visible that IA-TEAC and IA-TBAC DEMs polymerize much slower than the other DEMs. The polymerization of IA-CC, IA-BCC, IA-CCC and IA-AcCC DEMs slows down once the monomer conversion exceeds approximately 40 %, and finally reaches 60–70 % IA conversion. IA-TEAC and IA-TBAC DEMs are after 30 min still polymerize at an approximately constant (slow) rate.. Fig. 3b shows the IA conversion curves for individual IA-QCl DEMs photopolymerization at $[IA_0]$ = 4.8 mol/L. As in the previous case ($[IA_0]$ = 3.8 mol/L), IA-TEAC DEM polymerize significantly slower than the other DEMs. IA-CC, IA-CCC, IA-AcCC and IA-TMAC DEMs polymerize with similar rates, but in the case of IA-TMAC, the polymerization slows down and stops at a lower monomer conversion.

None of the studied IA-QCl DEMs reach 100 % monomer conversion after 30 min of photopolymerization (Fig. 3 and Tab. S3) due to DEM vitrification. Similar effect is observed for bulk polymerization of vinyl monomers [46], acrylic DEMs [47] and polymerizable ionic liquids [48,49]. When the glassy state is reached, the polymerization reaction is practically stopped due to the limited mobility of the monomer molecules and the macroradicals, which prevents further propagation [35]. This “glass-effect” phenomenon occurs for IA-CC, IA-BCC, IA-AcCC, IA-CCC and IA-TMAC DEMs. However, the IA-TEAC and IA-TBAC DEMs do not vitrify after 30 min of photopolymerization reaching a low final monomer conversion due to slow polymerization rate.

Fig. 4a shows the values of initial polymerization rate (R_p) for all IA-QCl DEMs. Due to the large difference in R_p, DEMs can be divided into two groups – *fast polymerizing* ($R_p > 2.5$ mmol·L^{-1}·s^{-1}) - IA-CC, IA-BCC, IA-CCC, IA-AcCC, IA-TMAC) and *slow polymerizing* ($R_p < 1.5$ mmol·L^{-1}·s^{-1}) - IA-AcCC, IA-TEAC and IA-TBAC). Clearly, the length of QCl alkyl chain is crucial for IA polymerization rates in IA-QCl DEMs, where increasing alkyl size causes the IA polymerization rate to significantly decrease in the order IA-TMAC ≫ IA-TEAC > IA-TBAC.

Fig. 4b shows that the average molecular weight (M_w) of the resulting PIA strongly depends on both the QCl type and $[IA_0]$ varying from low (M_w = 16 000 g/mol) to high values (M_w = 300 000 g/mol). According to the theory of radical polymerization kinetics, an increase in $[IA_0]$ should result in a higher degree of polymerization, which is observed for IA-TBAC, IA-TEAC, IA-BCC, IA-AcCC and in a lesser extent for IA-CC. In general, the polymerizations performed in IA-TEAC and IA-TBAC lead to a lower molecular weight PIA than in the other DEMs, similar to our earlier results [33]. However, even polymerization in IA-TBAC DEM, which is the least favourable for obtaining high molecular weight PIA, allows formation of PIA with a higher molecular weight ($M_w > 100\,000$ g/mol) than polymerization in water or organic solvents [13,14].

Summing up, it is clear that the polymerization process varies greatly among the investigated DEMs systems. Therefore, in the next part of the paper, we will try to clarify the relationship between the polymerization parameters and physicochemical properties of IA-QCl DEMs (Tab. S2). It is important to mention that the physicochemical properties of the DEMs are only determined for the initial polymerization times and a slight change is presumed with the progress of the IA polymerization.

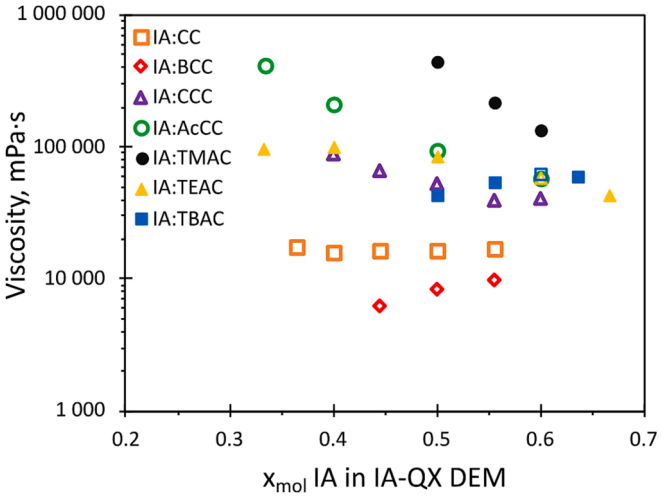


Fig. 5. Semi-logarithmic plot of viscosity of IA-QCl DEMs at 22 °C.

3.3. Effect of DEMs viscosity on IA polymerization

The viscosities of IA-QCl DEMs vary in the range of 6 000–450 000 mPa·s (Fig. 5), which is typical for DES formed by dicarboxylic acids [41,50]. According to the hole theory of liquids, the viscosity of DES decreases with increasing free volume [42,51]. Therefore, the viscosity of the IA-TMAC DEM is higher than that of DEMs formed by larger QCls (IA-TEAC and IA-TBAC). IA-CC and IA-BCC DEMs containing QCls with hydroxyl groups have significantly lower viscosities (6 000 – 17 000 mPa·s) than the other DEMs (40 000 – 450 000 mPa·s). BCC containing two hydroxyl groups forms a least viscous DEM (6 000 – 10 000 mPa·s) contrary to CC-based DEM (17 000 mPa·s) containing one hydroxyl group and IA-AcCC DEM (60 000 – 425 000 mPa·s) in which the hydroxyl group is esterified. These results indicate that the presence of free hydroxyl groups on the ammonium salt cation tend to reduce DEM viscosity.

Fig. 6a shows the ^{13}C NMR spectra of all IA-QCl DEMs (1:1 mol ratio) with respective structures and assignment of the carbon signals measured under solvent-free conditions at room temperature (22 °C). The resulting spectra show clear correlation between increased viscosity and broadening of the carbon signals as it is known that increased viscosity shortens the lifetime of excited nuclear magnetic states and causes broadening of NMR signals [52].

The sharp and distinguishable NMR signals in IA-CC and IA-BCC indicate sufficient mobility of the mixture components. This is consistent with rheological results showing significantly lower viscosity of IA-CC and IA-BCC systems as compared to the other DEMs (Fig. 5). Fig. 6b shows the correlation of IA methylene carbon FWHMs with macroscopic DEMs shear viscosity further proving that the increase in DEMs viscosity is related to the decrease in IA mobility in DEMs. For quantitative characterization of DEMs, ^{13}C NMR carbon signals full width at half-maximum (FWHM) analysis is provided (Tab. S4) [53,54]. Analysis of ^{1}H NMR spectra provides similar resuts (Fig. S4).

It is well known that radical polymerization is a process strongly affected by the viscosity of the reaction system [35,55] where high viscosity controls the termination reactions by limiting the mobility of the propagating macroradicals resulting in an accelerated polymerization rate and an increase in the molecular weight of the produced polymer. This phenomenon is termed the gel or Trommsdorff–Norrish effect [35]. For ionic liquids, which resemble DESs in terms of chemical structure and physicochemical properties, it is well-established that the

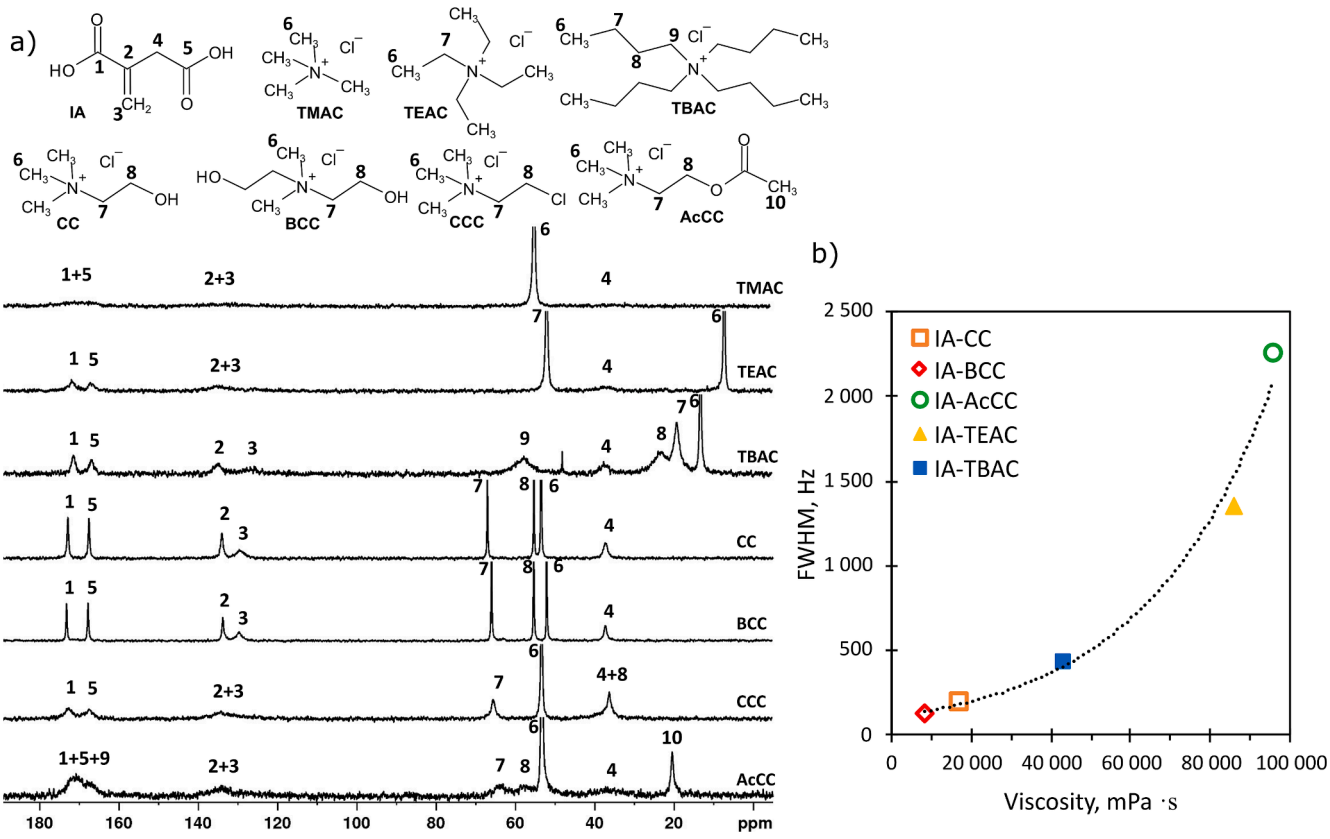


Fig. 6. A) ^{13}C NMR IA-QCl DEMs (1:1 M ratio) measured at 22 °C, b) Exponential correlation between viscosity of IA-QCl DEMs (1:1 M ratio) and ^{13}C NMR signal full width at half-maximum (FWHM) of methylene group of IA in DEMs at room temperature – the dotted line is only for leading the eyes. Note: IA-TMAC and IA-CCC DEMs were excluded from the plot due to the barely visible signals of IA carbons or overlapping methylene signals, respectively.

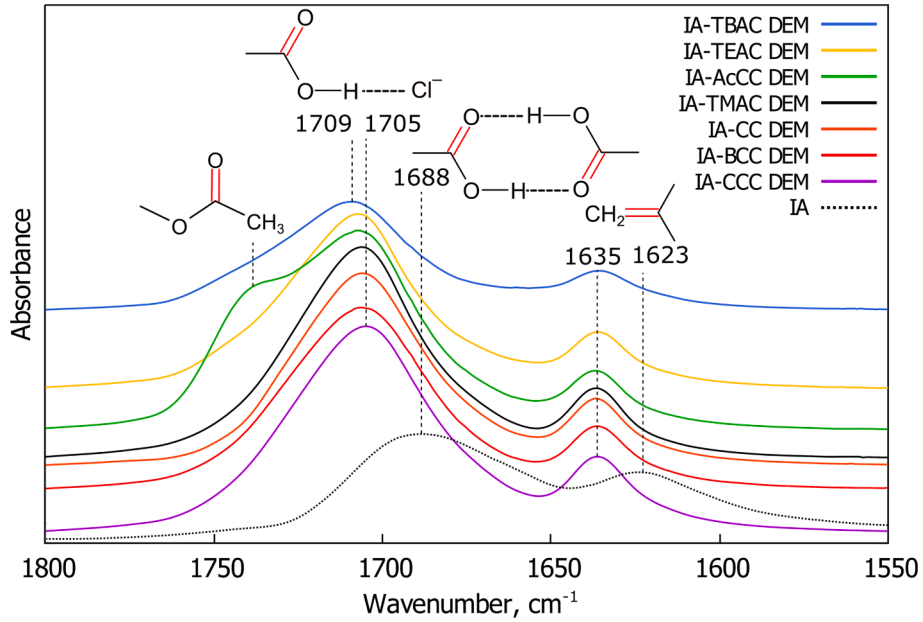


Fig. 7. Carbonyl region of FT-IR absorption spectra of IA-QCl DEMs (1:1 mol ratio) and IA with band assignment (bonds marked in red). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

viscosity has a significant effect on the polymerization kinetics [55]. In addition, this gel effect was also considered to be responsible for obtaining high molecular weight PIA in DEMs [15]. In contrast, in this study correlation between the initial viscosity of the DEMs and polymerization rate of IA and molecular weight of the polymers obtained was not observed (Fig. S5). Interestingly, PIA with the highest molecular weight was obtained in the lowest viscous IA-CC and IA-BCC DEMs. This may be related to the fact that when the viscosity of the reaction system is too high from the beginning, then not only the rate of termination, but also the propagation step is reduced resulting in decreased molecular

weight of the produced polymer [35].

3.4. Effect of the molecular state of IA on its polymerization

FT-IR spectroscopy allows determination of the state of IA in the DEMs through analysis of the band shift of the carbonyl group ($\nu_{(C=0)} \approx 1700\ cm^{-1}$). Fig. 7 shows the comparison of the IA-QCl 1:1 DEMs FT-IR spectra (1800–1600 cm^{-1}) in the ranges specific for the bands of carbonyl ($\nu_{(C=0)} \approx 1700\ cm^{-1}$) and vinyl ($\nu_{(C=C)} = 1635\ cm^{-1}$) groups. IA in its crystalline state exhibits chain interactions formed through hydrogen bonds between carboxyl groups, which are related to $\nu_{(C=0)\ IA} = 1688\ cm^{-1}$ [40]. As a result of IA-QCl DEMs formation, IA-IA interactions ($\nu_{(C=0)\ IA} = 1688\ cm^{-1}$) break, being replaced by IA-QCl complexes ($\nu_{(C=0)\ IA\text{-}QX} = 1705–1709\ cm^{-1}$), similarly as in the case of acrylic (AA) and methacrylic acids (MAA) DEMs, where monomers exist as dimers and monomer-QX complexes which remain in equilibrium, indicated in the FT-IR spectrum by a bimodal band of the carbonyl group [29,56]. However, in the case of IA-QCl DEMs, the carbonyl group band of IA is monomodal suggesting that it originates from a single form of IA associate, most likely the IA-QCl complex. Varying the molar ratio of the IA-QCl components of DEMs shows no change in the shape or width of the carbonyl group band (Fig. S6-12) suggesting that the molecular state of IA is not affected by the DEMs composition. Furthermore, the absence of the carboxylate band ($\nu_{(C=0)}$ IA = 1600–1550 cm^{-1}) confirms that IA in IA-QCl DEMs remains in a neutral, undissociated form [57]. This fact is important from the standpoint of the acid polymerization since the ionized IA is much less reactive in this process [11].

The Nile Red (NR) was used to investigate polarity of IA-QCl DEMs [36]. The shift of the NR absorbance maximum wavelength (λ_{max}) towards the longer λ values indicates the higher polarity of the system, represented as transition energy (E_{NR}). IA-CC, IA-BCC, IA-CCC and IA-TMAC DEMs have an approximately the same polarity ($E_{NR} \approx 47.7$), regardless of the molar composition. This makes these systems slightly more polar than water ($E_{NR} = 48.2$) [38]. The polarity of IA-AcCC (E_{NR} = 50.7–47.7), IA-TEAC (E_{NR} = 50.5–47.7) and IA-TBAC (E_{NR} = 49.7–47.9) DEMs is related to the type of QCl and increases with IA content.

To determine the interactions responsible for the polarity of a given system, a three-parameter Kamlet-Taft scale can be used, which consists of hydrogen-bond donating ability (α), hydrogen-bond accepting ability (β) and dipolarity/polarizability (π*) [58]. The process of radical polymerization can be affected by intermolecular interactions such as hydrogen bonding, electron pair donor/acceptor as well as non-specific interactions between solvent, monomer and propagating macroradicals [59] The solvatochromic Kamlet-Taft parameters have been successfully applied to describe the solvent effect on radical polymerization of vinyl monomers in ionic liquids [60–62].

For acidic systems, Reichardt's dye (Betaine 30) cannot be used to calculate α Kamlet-Taft parameter [58], therefore for IA-QCl DEMs only the β and π* parameters were determined (Tab. S3). The ability to non-specific interactions is described by π* and it was found to be similar for all investigated IA-QCl DEMs reaching high values in the range of 1.06–1.18 typical for water and ionic liquids [63]. It is known that the π* value of DES is mainly influenced by the HBD structure, while the type of HBA is of minor importance [58]. Since IA is the common HBD for all DEMs investigated, the determined π* values are very similar to each other.

The β parameter describing hydrogen bond accepting ability of DESs is mainly dependent on the type of HBA used [58]. For IA-QCl systems, chloride anion is mainly responsible for the formation of the hydrogen bonds with carboxylic groups of IA and is more prone to accept protons when it interacts weakly with the ammonium cation [58]. Thus, the strength of hydrogen bonding between IA and chloride ion can be tuned through the structure of the counterion [64]. In other words, the greater the β value, the weaker Q^+-Cl^- ionic interactions while at the same time the stronger H-bonding are formed between chloride anion and the acid.

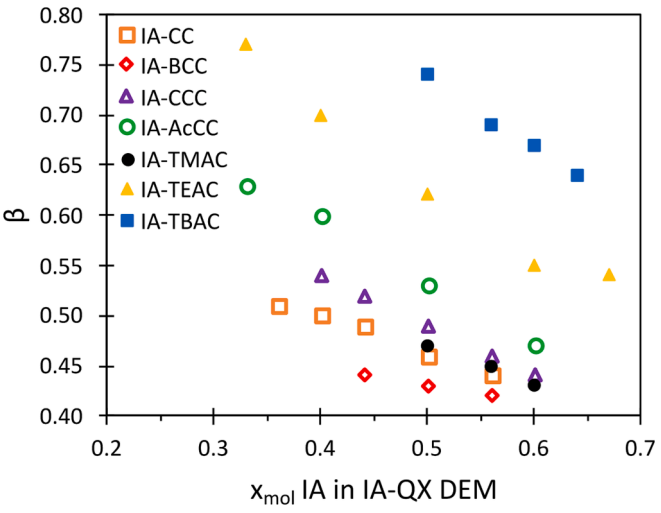


Fig. 8. Hydrogen bond accepting ability (β) of IA-QCl DEMs at room temperature.

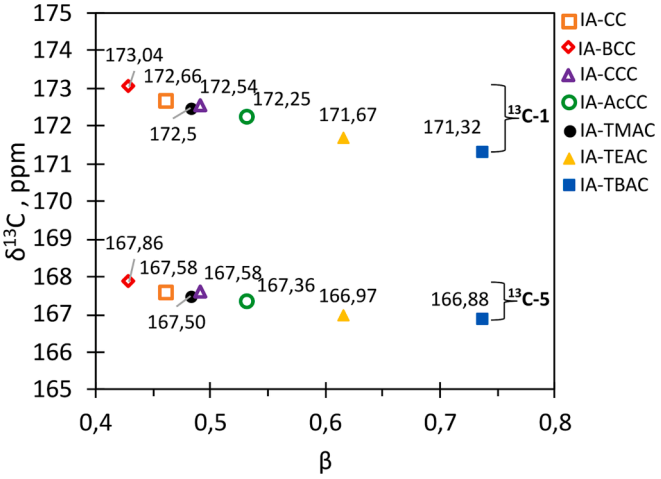


Fig. 9. Correlation between $\delta^{13}C$ NMR of IA carboxylic carbons (C-1 and C-5 according to assignment on Fig. 6a) in IA-QCl DEMs (1:1 M ratio) and β of IA-QCl DEMs (1:1 M ratio) measured at 60 °C in order to obtain spectra of better resolution (Fig. S14). Note, the measurement temperature does not cause significant changes in $\delta^{13}C$ (Fig. S15, S17).

For all DEMs, β increases with increasing QCl content (Fig. 8). The type of QCl affects β as well, especially in the alkylammonium salts series: IA-TMAC < IA-TEAC < IA-TBAC. As the QCl alkyl chain length increases, β increases significantly. This is a general trend observed for other DESs related to the size of the ammonium ion [58,65]. As the size of the cation increases, its charge density decreases, resulting in a weaker interaction with the chloride anion. This explains why the anion becomes more prone to accept protons from HBD [66]. The observed lowest values of β in the case of IA-CC and IA-BCC DEMs may be related to the presence of hydroxyl groups located on the ammonium cations. These additional polar sites could participate to the formation of H-bonds both with carboxylic groups and chloride anion making the system less susceptible to form additional hydrogen bonds with the solute molecules (e.g. a probe dye) [67,68].

Carboxylic carbons chemical shift ($\delta^{13}C$) analysis allows further clarification of hydrogen bond accepting properties of DEMs [29]. It has been reported that the ability to form hydrogen bonds in imidazole ionic liquids can be represented as a linear correlation between the chemical shift of the most acidic proton and the β parameter of ionic liquids

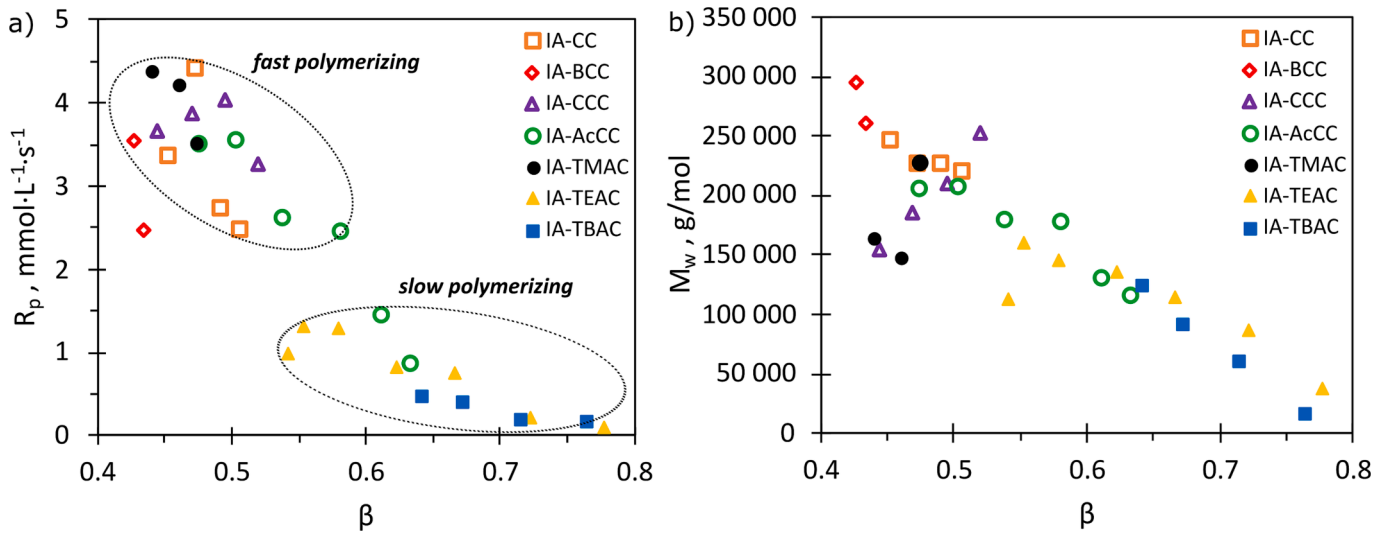


Fig. 10. Correlation of IA-QCl DEMs hydrogen bond accepting ability (β) with: a) polymerization rate; b) M_w of the obtained PIA.

[69,70]. In a similar way, Fig. 9 shows a strong correlation between the $\delta^{13}C$ of both carboxylic carbons (C-1 and C-5) and β of the IA-QCl DEMs.

At the same time, although not as clearly visible, there is an upfield-shift of $\delta^{13}C$ of the carboxylic carbons following the increasing β parameter. Typically, the carbon at the hydrogen bond donating group is unshielded and its $\delta^{13}C$ is higher the stronger the hydrogen bond [71–73]. Upon formation of DES, $\delta^{1}H$ and $\delta^{13}C$ upfield-shift of hydrogen bond donating group occurs caused by the weakening of the original hydrogen bonding network between HBD molecules through its replacement by HBD-HBA interactions [29,53]. In a similar manner, an up-field shift of $\delta^{31}P$ is observed in phosphinic acid organic derivatives as a result of hydrogen bond formation with HBA.[74,75] Furthermore, the $\delta^{13}C$ of the carboxylic carbon in HBD is additionally shifted down-field when its carbonyl group participates in hydrogen bond formation as the hydrogen bond accepting group [76]. Since the quaternary ammonium cation is a weaker hydrogen bond donor group than the carboxylic acid, it can be presumed that the carboxylic carbon up-field shift upon IA-QX DEMs formation is also related to the weaker contribution of the carbonyl group to the hydrogen bond formation [77]. On the other hand, full explanation of the carboxylic carbon up-field shift phenomenon due to hydrogen bonding interaction with QCls would require dedicated NMR studies and is out of scope of this paper.

Clearly, the β parameter reflects the complexity of the HBD-HBA interactions in the DEM. Therefore, it seems to be an ideal tool for elucidating the effect of QCl on IA polymerization. Fig. 10a shows correlation between R_p and β. Similarly to Fig. 4a, IA-QCl DEMs could be divided into *fast polymerizing* ($R_p > 2.5$ mmol·L^{-1}·s^{-1} and lower β) and *slow polymerizing* DEMs ($R_p < 1.5$ mmol·L^{-1}·s^{-1} and higher β). Fig. 10b shows an almost linear correlation between decreasing M_w of PIA and increasing β factor of IA-QCl DEM. The β value depends on both the type and the concentration of QCl in DEM, and allows for predicting the influence of IA-QCl DEM on the course of IA polymerization.

Explanation for the increasing hydrogen bond accepting ability (the increasing β) of the DEM system which makes IA less prone to polymerize is still not clear. It can be suspected that as a result of strong hydrogen bonding with QCl, the protons of the IA carboxyl groups are pulled away by chloride, resulting in the generation of a partial negative charge on the carboxylic groups. These forms of the monomer may propagate more slowly due to electrostatic repulsion [78]. The above tentative explanation is supported by the results of studies on several DES systems indicating that as the size of the QCl cation increases, charge delocalization due to chloride anion transfer is also increased [79,80,64,81].

4. Conclusions

In this study, the effect of the quaternary ammonium chlorides (QCl) cation structure on radical polymerization of itaconic acid (IA) in IA-QCl deep eutectic monomers (DEMs) was elucidated. QCls cation structure affects both the physicochemical properties (density, viscosity and hydrogen-bond accepting ability – β) of the IA-QCl DEMs and the IA polymerization process – polymerization rate and molecular weight of the obtained PIA.

The β parameter reflects the hydrogen bonding between IA and QCls while being dependent simultaneously on both the type and the concentration of QCl in DEM. It has been found that β of the IA-QCl DEMs is inversely correlated to IA polymerization rate and the molecular weight of the resulting PIA. Thus, the value of β allows prediction of the QCls cation structure and its effect on the progress and outcome of IA radical polymerization in IA-QCl DEMs.

In conclusion, the use of IA-QCl DEMs comprising a weak hydrogen bond acceptor QCl – i.e. bis(2-hydroxyethyl)dimethylammonium chloride (BCC) or choline chloride (CC) enables controlled PIA synthesis reaching otherwise impossibly high molecular weights of PIA, which is a fundamental requirement for polymer material design and further applications.

CRediT authorship contribution statement

Kacper Mielczarek: . **Samuel Wierzbicki:** . **Monika Topa-Skwarczyńska:** . **Sonia Bujok:** . **Rafał Konefał:** . **Martina Nevoralová:** Investigation. **Joanna Ortyl:** . **Hynek Beneš:** . **Szczepan Bednarz:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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